Rosen — §5.3: Recursive Definitions and Structural Induction

Selected Exercises

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Exercise 14

In this and the next exercise, f_n is the nth Fibonacci number.
Show that $f_{n+1}f_{n-1} - f_n^2 = (-1)^n$ when n is a positive integer.

Exercise 18

Let

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

Show that

$$A^n = \begin{bmatrix} f_{n+1} & f_n \\ f_n & f_{n-1} \end{bmatrix}$$

when n is a positive integer.

Exercise 20

Give a recursive definition of the functions max and min so that $\max(a_1, a_2, \ldots, a_n)$ and $\min(a_1, a_2, \ldots, a_n)$ are the maximum and minimum of the n numbers $a_1, a_2, \ldots, a_n$, respectively.

Exercise 21

Let $a_1, a_2, \ldots, a_n$, and $b_1, b_2, \ldots, b_n$ be real numbers. Use the recursive definitions that you gave in Exercise 20 to prove these.

(a) $\max(-a_1, -a_2, \ldots, -a_n) = -\min(a_1, a_2, \ldots, a_n)$

(b) $\max(a_1 + b_1, a_2 + b_2, \ldots, a_n + b_n) \leq \max(a_1, a_2, \ldots, a_n) + \max(b_1, b_2, \ldots, b_n)$

(c) $\min(a_1 + b_1, a_2 + b_2, \ldots, a_n + b_n) \geq \min(a_1, a_2, \ldots, a_n) + \min(b_1, b_2, \ldots, b_n)$

Exercise 26

Let S be the subset of the set of ordered pairs of integers defined recursively by
Basis step: $(0, 0) \in S$.
Recursive step: If $(a, b) \in S$, then $(a + 2, b + 3) \in S$ and $(a + 3, b + 2) \in S$.

(a) List the elements of S produced by the first five applications of the recursive definition.

(b) Use strong induction on the number of applications of the recursive step of the definition to show that $5 \mid a + b$ when $(a, b) \in S$.

(c) Use structural induction to show that $5 \mid a + b$ when $(a, b) \in S$.

Exercise 32

(a) Give a recursive definition of the function $ones(s)$, which counts the number of ones in a bit string s.

(b) Use structural induction to prove that $ones(st) = ones(s) + ones(t)$.

Exercise 36

The reversal of a string is the string consisting of the symbols of the string in reverse order. The reversal of the string w is denoted by w^R.

Use structural induction to prove that $(w_1w_2)^R = w_2^R w_1^R$.

Exercise 38

Give a recursive definition of the set of bit strings that are palindromes.

Exercise 44

The set of leaves and the set of internal vertices of a full binary tree can be defined recursively.

Basis step: *The root r is a leaf of the full binary tree with exactly one vertex r. This tree has no internal vertices.*

Recursive step: *The set of leaves of the tree $T = T_1 \cdot T_2$ is the union of the sets of leaves of T_1 and of T_2. The internal vertices of T are the root r of T and the union of the set of internal vertices of T_1 and the set of internal vertices of T_2.*

Use structural induction to show that $l(T)$, the number of leaves of a full binary tree T, is 1 more than $i(T)$, the number of internal vertices of T.

Exercise 47

A *partition* of a positive integer n is a way to write n as a sum of positive integers where the order of terms in the sum does not matter. For instance, $7 = 3 + 2 + 1 + 1$ is a partition of 7. Let P_m equal the number of different partitions of m, and let $P_{m,n}$ be the number of different ways to express m as the sum of positive integers not exceeding n.

(a) Show that $P_{m,m} = P_m$.

(b) Show that the following recursive definition for $P_{m,n}$ is correct:

$$P_{m,n} = \begin{cases} 1 & \text{if } m = 1 \\ 1 & \text{if } n = 1 \\ P_{m,m} & \text{if } m < n \\ 1 + P_{m,m-1} & \text{if } m = n > 1 \\ P_{m,n-1} + P_{m-n,n} & \text{if } m > n > 1. \end{cases}$$

(c) Find the number of partitions of 5 and of 6 using this recursive definition.

Exercise 58

Show that each of these proposed recursive definitions of a function on the set of positive integers does not produce a well-defined function.

(a) $F(n) = 1 + F(\lfloor n/2 \rfloor)$ for $n \geq 1$ and $F(1) = 1$.

(b) $F(n) = 1 + F(n-3)$ for $n \geq 2$, $F(1) = 2$, and $F(2) = 3$.

(c) $F(n) = 1 + F(n/2)$ for $n \geq 2$, $F(1) = 1$, and $F(2) = 2$.

(d) $F(n) = 1 + F(n/2)$ if n is even and $n \geq 2$, $F(n) = 1 - F(n-1)$ if n is odd, and $F(1) = 1$.

(e) $F(n) = 1 + F(n/2)$ if n is even and $n \geq 2$, $F(n) = F(3n-1)$ if n is odd and $n \geq 3$, and $F(1) = 1$.

Exercise 53

Recall Ackermann's function:

$$A(m,n) = \begin{cases} 2n & \text{if } m = 0 \\ 0 & \text{if } m \geq 1 \text{ and } n = 0 \\ 2 & \text{if } m \geq 1 \text{ and } n = 1 \\ A(m-1, A(m, n-1)) & \text{if } m \geq 1 \text{ and } n \geq 2. \end{cases}$$

Prove that $A(m, n+1) > A(m, n)$ whenever m and n are nonnegative integers.